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SSC-I NATIONAL BOOK FOUNDATION FBISE

CHEMISTRY TEST - CHAPTER 16: EMPIRICAL DATA COLLECTION AND ANALYSIS

Name:		Roll No.		Section:

Total Time: 2:30 Hour
Time Allowed: 150 Minutes

Total Marks: 150

PAPER OBJECTIVE
SECTION-A (MARKS 150)

NOTE: Section-A is compulsory. All parts of this section are to be answered on these pages and handed over to the center superintendent. Deleting / overwriting is not allowed. Do not use lead pencil.

Q.1 Fill the relevant bubble for each part. All parts carry one mark.

Questions	A	B	C	D	A B C D
1. Measurement is a comparison of an unknown quantity with a known fixed:	Name	Colour	Shape	Quantity	○○○●
2. Normal human body temperature is:	110 °F	98.6 °F	102 °F	90 °F	○●○○
3. A body temperature of 102 °F indicates:	Low temperature	Fever	Normal health	Death	○●○○
4. Accurate and precise measurement is a fundamental component of:	Chemistry	Sports only	Fashion	Music	●○○○
5. In the past, people used abstract units like the cubit or:	Litre	Metre	Kilogram	Hand span	○○○●
6. The cubit or hand span varied from person to person, making old measurements:	Perfect	Inaccurate	Standard	Precise	○●○○
7. SI stands for:	Simple Indicator	Standard Index	Scientific Instrument	International System of Units	○○○●
8. The International System of Units was adopted in:	1960	2000	1860	1760	●○○○
9. In chemistry, SI units are used to measure mass, volume and:	Colour	Taste	Smell	Temperature	○○○●
10. A quantity that can be measured is a:	Physical quantity	Symbol	Unit	Prefix	●○○○
11. Which of these is a physical quantity?	Beauty	Joy	Mass	Taste	○○●○
12. A fixed quantity used to compare unknown quantities is a:	Unit of measurement	Prefix	Symbol	Formula	●○○○

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Questions	A	B	C	D	A B C D
13. The standard unit of measurement is a value that is:	Fixed and cannot be changed	Random	Personal	Variable	●○○○
14. The metric system has _____ types of units.	Seven	Two	Ten	Three	○●○○
15. The two types of units in SI are fundamental units and:	Derived units	Mixed units	Prefix units	Local units	●○○○
16. The number of fundamental (base) SI units is:	Ten	Seven	Five	Three	○●○○
17. Which is a fundamental SI unit?	Pascal	Kilogram	Newton	Joule	○●○○
18. Another example: Which is a fundamental SI unit?	Metre	Joule	Newton	Pascal	●○○○
19. Yet another: Which is a fundamental SI unit?	Watt	Second	Joule	Newton	○●○○
20. One more: Which is a fundamental SI unit?	Joule	Pascal	Newton	Kelvin	○○○●
21. Also: Which is a fundamental SI unit?	Pascal	Joule	Mole	Newton	○○●○
22. The SI unit of the amount of substance is the:	Mole	Litre	Gram	Kelvin	●○○○
23. The SI unit of temperature is the:	Celsius	Kelvin	Calorie	Fahrenheit	○●○○
24. The SI unit of mass is the:	Pound	Gram	Tonne	Kilogram	○○○●
25. The SI unit of length is the:	Metre	Foot	Centimetre	Inch	●○○○
26. The SI unit of time is the:	Second	Minute	Day	Hour	●○○○
27. Units composed of two or more fundamental units are called:	Derived units	Prefix units	Standard units	Base units	●○○○
28. Derived units are:	Fundamental	Not independent	Independent	Fixed	○●○○
29. The SI unit of volume is the:	Millilitre	Cubic metre	Litre	Cubic centimetre	○●○○

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Questions	A	B	C	D	A B C D
30. SI prefixes are used to identify the _____ or fractions of the original unit.	Masses	Colours	Multiples	Names	○○●○
31. 1 decametre (dam) equals:	100 m	1000 m	10 m	1 m	○○●○
32. 1 hectometre (hm) equals:	100 m	1 m	1000 m	10 m	●○○○
33. 1 kilometre (km) equals:	10000 m	10 m	1000 m	100 m	○○●○
34. 1 decimetre (dm) equals:	10 m	1 m	0.1 m	0.01 m	○○●○
35. 1 centimetre (cm) equals:	0.001 m	0.1 m	1 m	0.01 m	○○○●
36. 1 millimetre (mm) equals:	0.1 m	0.001 m	1 m	0.01 m	○●○○
37. The prefix 'kilo' means:	10^{-3}	10^{2}	10^{3}	10^{6}	○○●○
38. The prefix 'milli' means:	10^{-3}	10^{3}	10^{-6}	10^{-2}	●○○○
39. The prefix 'micro' means:	10^{-3}	10^{6}	10^{-9}	10^{-6}	○○○●
40. The prefix 'nano' means:	10^{9}	10^{-3}	10^{-9}	10^{-6}	○○●○
41. The prefix 'mega' means:	10^{9}	10^{6}	10^{3}	10^{-6}	○●○○
42. The prefix 'giga' means:	10^{9}	10^{12}	10^{3}	10^{6}	●○○○
43. The prefix 'centi' means:	10^{-2}	10^{-3}	10^{-1}	10^{2}	●○○○
44. One nanometre equals:	10^{-3} m	10^{-9} m	10^{-6} m	10^{9} m	○●○○
45. A distance of 1 kilometre means:	1000 m	1000 cm	100 m	10 m	●○○○
46. 1 kg equals:	10000 g	10 g	100 g	1000 g	○○○●
47. 1 g equals:	10 mg	100 mg	10000 mg	1000 mg	○○○●
48. 12 kilograms equals _____ grams.	12,000	12	1,200	120	●○○○

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Questions	A	B	C	D	A B C D
49. 12 kilograms equals _____ milligrams.	12,000	1,200,000	12,000,000	120,000	○○●○
50. To convert from a larger unit to a smaller unit, we:	Subtract	Divide	Add	Multiply	○○○●
51. To convert from a smaller unit to a larger unit, we:	Subtract	Divide	Add	Multiply	○●○○
52. A nurse needs to give 0.2 g from 50 mg tablets. She needs _____ tablets.	5	2	10	4	○○○●
53. 0.2 g equals _____ mg.	2	2000	20	200	○○○●
54. 1 m equals _____ cm.	100	10	1	1000	●○○○
55. 1 cm equals _____ mm.	1	10	1000	100	○●○○
56. 1 litre equals _____ millilitres.	1000	10000	100	10	●○○○
57. 1 litre equals:	1 dm^3	10 dm^3	1 m^3	1 cm^3	●○○○
58. 1 dm^3 equals:	100 cm^3	1 cm^3	1000 cm^3	10 cm^3	○○●○
59. 1000 cm^3 equals:	100 L	10 L	1 litre	1 mL	○○●○
60. 1 mL equals:	1 cm^3	100 cm^3	10 cm^3	1000 cm^3	●○○○
61. Chemists use cm^3, g and s in the lab because they are more:	Practical for small amounts	Costly	Rare	Difficult	●○○○
62. Using grams rather than kilograms provides more:	Slow measurements	Wrong measurements	Precise measurements	Costly measurements	○○●○
63. Using seconds as the unit of time helps measure:	Colour	Density	Mass	Reaction rates	○○○●
64. Scientific notation is also called:	Fraction form	Decimal form	Standard form	Ordinary form	○○●○
65. Scientific notation is used for very large or very:	Round numbers	Even numbers	Small numbers	Odd numbers	○○●○
66. In scientific notation, a number is written in the form:	A − 10^n	A + 10^n	A ÷ 10^n	A × 10^n	○○○●

Questions	A	B	C	D	A B C D
67. In A × 10^n, the value of A satisfies:	1 ≤ A < 10	0 < A < 1	10 ≤ A < 100	A > 10	●○○○
68. In A × 10^n, n is:	Always positive	A fraction only	Always zero	An integer	○○○●
69. The base in scientific notation is always:	10	5	2	100	●○○○
70. The exponent is positive if the number is very:	Small	Even	Round	Large	○○○●
71. The exponent is negative if the number is very:	Round	Even	Large	Small	○○○●
72. 650,000,000 in scientific notation is:	65×10^7	6.5×10^7	6.5×10^8	6.5×10^9	○○●○
73. 490,000,000 in scientific notation is:	4.9×10^9	4.9×10^8	49×10^7	4.9×10^7	○●○○
74. 1,230,000,000 in scientific notation is:	1.23×10^{10}	1.23×10^8	1.23×10^9	12.3×10^8	○○●○
75. 50,500,000 in scientific notation is:	50.5×10^6	5.05×10^6	5.05×10^7	5.05×10^8	○○●○
76. 0.000000097 in scientific notation is:	9.7×10^{-7}	9.7×10^{-8}	9.7×10^8	97×10^{-9}	○●○○
77. 0.0000212 in scientific notation is:	2.12×10^{-5}	2.12×10^{-4}	21.2×10^{-6}	2.12×10^5	●○○○
78. 0.006 in scientific notation is:	6×10^{-2}	60×10^{-4}	6×10^{-3}	6×10^3	○○●○
79. 0.00000046 in scientific notation is:	4.6×10^{-6}	46×10^{-8}	4.6×10^7	4.6×10^{-7}	○○○●
80. 38,000,000,000 in scientific notation is:	3.8×10^9	3.8×10^{10}	3.8×10^{11}	38×10^9	○●○○
81. 5000 in scientific notation is:	5×10^3	50×10^2	5×10^4	0.5×10^4	●○○○
82. 300,000,000 in scientific notation is:	30×10^7	3×10^7	3×10^9	3×10^8	○○○●
83. 602,000,000,000,000,000,000,000 in scientific notation is:	6.02×10^{24}	6.02×10^{22}	6.02×10^{23}	60.2×10^{22}	○○●○

Questions	A	B	C	D	A B C D
84. The number 14 × 10³ is not in standard form because 14 is:	Negative	Greater than 10	A fraction	Less than 1	○●○○
85. 14 × 10³ in standard form is:	1.4 × 10³	1.4 × 10²	1.4 × 10⁴	14 × 10⁴	○○●○
86. If a number is smaller than 1, the decimal point is moved to the:	Up	Right	Down	Left	○●○○
87. If a number is larger than 10, the decimal point is moved to the:	Down	Right	Up	Left	○○○●
88. The power of 10 for numbers smaller than 1 is:	Negative	One	Positive	Zero	●○○○
89. The mantissa carries the rest of the _____ digits of the number.	Zero	Decimal	Significant	Rounded	○○●○
90. The coefficient in scientific notation can be:	Only zero	Only negative	Positive or negative	Only positive	○○●○
91. Accuracy is a measure of how close a measurement is to the:	Average only	Last value	True value	First value	○○●○
92. Precision refers to how close repeated measurements are to:	The true value	Each other	Zero	The mean only	○●○○
93. If a balance is not adjusted correctly, you can make precise but _____ measurements.	Accurate	Fast	Inaccurate	Few	○○●○
94. Accuracy is usually related to how well the equipment is:	New	Painted	Heavy	Adjusted	○○○●
95. Precision is usually determined by how _____ the scientist is.	Fast	Tall	Old	Careful	○○○●
96. A careless spill of a sample makes repeated measurements not:	Cold	Accurate	Precise	Heavy	○○●○
97. When reading a measuring cylinder, the liquid forms a curve called the:	Margin	Meniscus	Meridian	Miniscule	○●○○
98. The meniscus of water is a _____ shape.	U	Dome	V	Flat	●○○○
99. To read a measuring cylinder accurately, your eye should be:	At eye level with the meniscus	Below the meniscus	Far away	Above the meniscus	●○○○
100. If you read from above the meniscus, you read a _____ volume.	Smaller	Same	Zero	Larger	●○○○
101. If you read from below the meniscus, you read a _____ volume.	Zero	Smaller	Larger	Same	○○●○

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Questions	A	B	C	D	A B C D
102. Beakers measure _____ volume of liquids.	Fixed	Approximate	Zero	Exact	○●○○
103. Beakers can be used for holding samples, stirring and:	Cutting	Timing	Weighing	Heating liquids	○○○●
104. Measuring cylinders are more accurate than:	Balances	Burettes	Pipettes	Beakers	○○○●
105. Measuring cylinders are commonly found in sizes of 10, 25, 50 and:	5 L	100 cm^3	1 cm^3	1000 L	○●○○
106. Pipettes are designed to measure:	Any volume	Time	A single volume accurately	Mass	○○●○
107. The usual volume of a pipette is:	500 cm^3	1000 cm^3	1 cm^3	25 cm^3	○○○●
108. A pipette should only be used with a pipette:	Filler	Stand	Clamp	Boss	●○○○
109. Burettes deliver any volume up to:	5 cm^3	50 cm^3	1000 cm^3	500 cm^3	○●○○
110. Burettes are more accurate than measuring cylinders but less accurate than:	Syringes	Cups	Beakers	Pipettes	○○○●
111. Burettes are mainly used in:	Weighing	Titrations	Heating	Timing	○●○○
112. A burette needs a stand, clamp and:	Syringe	Boss	Balance	Filler	○●○○
113. The volume delivered from a burette is final minus _____ reading.	Initial	Average	Zero	Middle	●○○○
114. Gas syringes are used to collect and measure the volume of:	Gas	Liquid	Powder	Solid	●○○○
115. Gas syringes have similar accuracy to:	Burettes	Pipettes	Measuring cylinders	Balances	○○●○
116. A laboratory thermometer ranges from:	0 °C to 10 °C	−10 °C to 110 °C	50 °C to 60 °C	−50 °C to 500 °C	○●○○
117. The laboratory thermometer is used to check the boiling and _____ points.	Dew	Freezing	Burning	Melting of gold	○●○○
118. A balance is used in quantitative chemical analysis to determine:	Time	Mass	Temperature	Volume	○●○○

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Questions	A	B	C	D	A B C D
119. An electronic balance provides _____ results.	Analog	Digital	None	Verbal	○●○○
120. A traditional (mechanical) balance does not need:	Weights	A scale	A pan	Electricity	○○○●
121. A stopwatch measures:	Temperature	Volume	Time intervals	Mass	○○●○
122. The precision of most stopwatches is:	1 hour	1 second	0.1 minute	0.01 seconds	○○○●
123. Time interval is defined as the elapsed time between two:	Masses	Colours	Volumes	Events	○○○●
124. Experimental methods give researchers a high level of:	Cost	Confusion	Control	Delay	○○●○
125. A disadvantage of experimental methods is that they can be:	Quick	Time-consuming and costly	Cheap	Simple	○●○○
126. A disadvantage of experimental methods is that they can lead to _____ situations.	Free	Natural	Fast	Artificial	○○○●
127. An advantage of experimental methods is that they can be used in different:	Books	Rooms only	Fields	Places only	○○●○
128. Which scientist is mentioned for experiments in chemistry?	Marie Curie	Galileo	Einstein	Newton	●○○○
129. Another example: Which scientist is mentioned for experiments in chemistry?	Einstein	Antoine Lavoisier	Galileo	Newton	○●○○
130. Yet another: Which scientist is mentioned for experiments in chemistry?	Newton	Einstein	Robert Boyle	Galileo	○○●○
131. One more: Which scientist is mentioned for experiments in chemistry?	Linus Pauling	Newton	Galileo	Einstein	●○○○
132. Chemists use experiments to see if new _____ are effective.	Books	Cars	Drugs	Toys	○○●○
133. Prefixes are also used in naming binary molecular compounds, like 'di' meaning:	Three	One	Four	Two	○○○●
134. The prefix 'mono' in molecular compound names means:	Four	Three	One	Two	○○●○
135. The prefix 'tri' in molecular compound names means:	Three	Five	Two	Four	●○○○

Questions	A	B	C	D	A B C D
136. The prefix 'tetra' in molecular compound names means:	Three	Five	Six	Four	○○○●
137. The prefix 'penta' in molecular compound names means:	Six	Five	Four	Seven	○●○○
138. The prefix 'hexa' in molecular compound names means:	Seven	Eight	Six	Five	○○●○
139. CO_2 is named:	Carbide	Carbon dioxide	Carbonate	Carbon monoxide	○●○○
140. CO is named:	Carbide	Carbon monoxide	Carbonate	Carbon dioxide	○●○○
141. N_2O_5 is named:	Nitrogen oxide	Dinitrogen trioxide	Nitrous oxide	Dinitrogen pentoxide	○○○●
142. The unit before Empirical Data Collection and Analysis is:	Matter	Hydrocarbons	Stoichiometry	Biochemistry	○○○●
143. The unit after Empirical Data Collection and Analysis is:	Biochemistry	Separation Techniques	Matter	Hydrocarbons	○●○○
144. The decimetre cube (dm^3) equals:	1 litre	100 mL	1 mL	10 L	●○○○
145. Which measuring tool is the most accurate for a fixed volume?	Pipette	Beaker	Measuring cylinder	Gas syringe	●○○○
146. Which measuring tool is used to deliver variable volumes in titration?	Beaker	Syringe	Pipette	Burette	○○○●
147. Which apparatus is the least accurate for measuring liquid volume?	Burette	Beaker	Pipette	Measuring cylinder	○●○○
148. A 25 cm^3 pipette delivers _____ volume when emptied.	A varying	Any	A single fixed	No	○○●○
149. The correct reading position for a measuring cylinder is:	From the side with closed eyes	Above the cylinder	Eye level	Below the cylinder	○○●○
150. Which SI base unit is represented by 'kg'?	Temperature	Time	Length	Mass	○○○●

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